

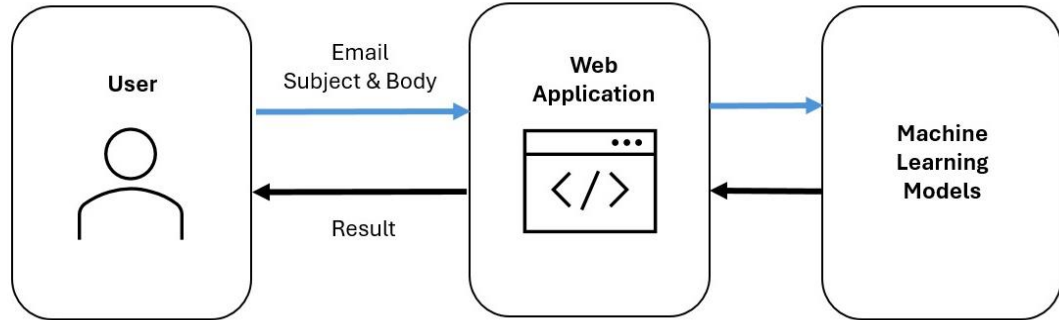
Security Analysis & Mitigations on UVIC Spam Detector

Group 11 MTIS



System Architecture

- **Stateless**
- **Lightweight**
- **No data retention**
- **Practical**



Security Analysis

- **Web Application**

- Man-in-the-Middle (MITM)
 - **Passive** (profiling Build databases of intercepted emails for later exploitation)
 - **Active** (Modify the response payload on the fly to manipulate classification results)
- Denial of Service (DoS/DDoS)

- **Machine Learning**

- Evasion
- Model Extraction

Mitigations

- **Rate Limiting & Monitoring**
 - Dos/DDos and Model Extraction
- **Output Minimization**
 - Evasion using feedback
- **HTTPS & HSTS Enforcement**
 - MITM Attack
- **AWS Service**
 - MITM and Dos/DDos

Future Extensions & Challenges

- Intrusion Detection Systems (IDS)
- Firewalls
- Email system APIs
- Browser extensions
- High accuracy
- Low latency
- Scalability

Thank You!



Work Division

- Zoe
 - Machine Learning Vulnerabilities & Mitigation
 - Report Drafting
- Lei
 - Web Application Vulnerabilities & Mitigation
 - Slides
- Peter
 - Conclusion
 - Proofreading and Revise